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| 17/134,756 | 12/28/2020 | SIDDHARTH RAVICHANDRAN | 200397-US-ORG1 | 3046 |

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| EXAMINER | |
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| BOYLE, ABBIGALE A | |

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**Please find below and/or attached an Office communication concerning this application or proceeding.**

The time period for reply, if any, is set in the attached communication.

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| Office Action Summary | Application No. 17/134,756 | Applicant(s) RAVICHANDRAN et al. | |
|---|---|---|---|
| | Examiner ABBIGALE A BOYLE | Art Unit 2899 | AIA (FITF) Status Yes |

*-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --*

**Period for Reply**

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

**Status**

1)☑ Responsive to communication(s) filed on 14 May 2024.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☑ This action is **FINAL.** 2b) ☐ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

**Disposition of Claims***

5) ☑ Claim(s) 9-10,12,23-29 and 36-45 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 9-10,12,23-29 and 36-45 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

**Application Papers**

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 28 December 2020 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

**Priority under 35 U.S.C. § 119**

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

**Certified copies:**

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

**Attachment(s)**

1) ☑ Notice of References Cited (PTO-892)

2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

## DETAILED ACTION

### *Notice of Pre-AIA or AIA Status*

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

### *Claim Rejections - 35 USC § 112*

The following is a quotation of the first paragraph of 35 U.S.C. 112(a):

> (a) IN GENERAL.—The specification shall contain a written description of the invention, and of the manner and process of making and using it, in such full, clear, concise, and exact terms as to enable any person skilled in the art to which it pertains, or with which it is most nearly connected, to make and use the same, and shall set forth the best mode contemplated by the inventor or joint inventor of carrying out the invention.

The following is a quotation of the first paragraph of pre-AIA 35 U.S.C. 112:

> The specification shall contain a written description of the invention, and of the manner and process of making and using it, in such full, clear, concise, and exact terms as to enable any person skilled in the art to which it pertains, or with which it is most nearly connected, to make and use the same, and shall set forth the best mode contemplated by the inventor of carrying out his invention.

Claims 9 and 36 are rejected under 35 U.S.C. 112(a) or 35 U.S.C. 112 (pre-AIA), first paragraph, as failing to comply with the written description requirement. The claim(s) contains subject matter which was not described in the specification in such a way as to reasonably convey to one skilled in the relevant art that the inventor or a joint inventor, or for applications subject to pre-AIA 35 U.S.C. 112, the inventor(s), at the time the application was filed, had possession of the claimed invention. The claims include the limitation of "wherein a mold compound encapsulates the system on a chip, the photonic integrated circuit, and the optical fiber…wherein the mold compound and the substrate include a cutout at the attachment of the optical fiber to the photonic integrated circuit".

However, in the embodiment of Figures 1a/1b, the mold compound does not encapsulate the optical fiber, as there is a cutout at the attachment of the optical fiber in the mold compound and the substrate.

Claims 25 and 41 are rejected under 35 U.S.C. 112(a) or 35 U.S.C. 112 (pre-AIA), first paragraph, as failing to comply with the written description requirement. The claim(s) contains subject matter which was not described in the specification in such a way as to reasonably convey to one skilled in the relevant art that the inventor or a joint inventor, or for applications subject to pre-AIA 35 U.S.C. 112, the inventor(s), at the time the application was filed, had possession of the claimed invention. The claims include the limitation of "wherein the substrate comprises a plurality of polymer layers". There is no disclosure of the substrate comprising polymer layers in the originally filed Application; there is only disclosure of the organic redistribution layer comprising multiple polymer layers.

### *Claim Rejections - 35 USC § 103*

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

**Claims 9-11, 23, 25, 28-29, 36-37, 39, 41, and 44-45 are rejected under 35 U.S.C. 103 as being unpatentable over Tai et al. (U.S. 2021/0157052) in view of Chen et al. (U.S. 2020/0271860).**

**Regarding Claim 9,** Tai et al. discloses an apparatus with optical fiber attach capability, the apparatus comprising:

a substrate (substrate 122, Figure 44);

a system on a chip disposed on the substrate (chip 50a, Figure 44);

a photonic integrated circuit with a v-groove in a front side fiber coupling region (PIC 50c, groove 76, Figure 44);

an organic redistribution layer communicating with the chip and photonic integrated circuit (organic redistribution layer 106, Figure 44); and

an optical fiber attached to the front side fiber coupling region (optical fiber 168, Figure 44),

wherein a mold compound encapsulates the system on a chip, the photonic integrated circuit, and the optical fiber (mold compound 120, Figure 44), and

wherein the substrate includes a cutout at the attachment of the optical fiber to the photonic integrated circuit (substrate 122, Figure 44).

However, they do not explicitly disclose that the mold compound includes a cutout at the attachment of the optical fiber to the photonic integrated circuit. Chen et al. discloses a similar device wherein a mold compound encapsulates a chip, a photonic integrated circuit with a groove in a front side, and an optical fiber, said the mold compound including a cutout at the attachment of the optical fiber to the photonic integrated circuit (chip 120, photonic integrated circuit 110c, optical fiber 160c, mold compound 130, Figure 15). Therefore it would have been obvious to one of ordinary skill in the art at the time of the invention to include a cutout in the mold compound at the attachment of the optical fiber to the photonic integrated circuit in Tai et al. in view of Chen

et al. in order to simplify the complexity of manufacturing and reduce production cost (Chen et al., Paragraph 38).

Regarding Claim 10, Tai et al. in view of Chen et al. further disclose wherein the apparatus is a die-last wafer- level fanout package (Tai et al., package 601, Figure 44). The language, term, or phrase "die-last wafer-level fanout package," is directed towards the process of making a wafer-level fanout package. It is well settled that "product by process" limitations in claims drawn to structure are directed to the product, per se, no matter how actually made. In re Hirao, 190 USPQ 15 at 17 (footnote 3). See also, In re Brown, 173 USPQ 685; In re Luck, 177 USPQ 523; In re Fessmann, 180 USPQ 324; In re Avery, 186 USPQ 161; In re Wethheim, 191 USPQ 90 (209 USPQ 554 does not deal with this issue); In re Marosi et al., 218 USPQ 289; and particularly In re Thorpe, 227 USPQ 964, all of which make it clear that it is the patentability of the final product per se which must be determined in a "product by process" claim, and not the patentability of the process, and that an old or obvious product produced by a new method is not patentable as a product, whether claimed in "product by process" claims or otherwise. The above case law further makes clear that applicant has the burden of showing that the method language necessarily produces a structural difference. As such, the language "die-last wafer-level fanout package" only requires a wafer-level fanout package with a die, which does not distinguish the invention from Tai et al. in view of Chen et al., who teaches the structure as claimed.

Regarding Claim 23, Tai et al. in view of Chen et al. further disclose wherein the mold compound comprises an epoxy material (Tai et al., mold compound 120, Figure 44, Paragraph 36).

Regarding Claim 25, Tai et al. in view of Chen et al. further disclose wherein the substrate comprises a plurality of polymer layers (Tai et al., substrate 122, Figure 44, Paragraphs 42).

Regarding Claim 28, Tai et al. in view of Chen et al. further disclose wherein a plurality of bumps is attached to the substrate (Tai et al., bumps 172 or 66, Figure 44).

Regarding Claim 29, Tai et al. in view of Chen et al. further disclose wherein the plurality of bumps comprises a ball grid array (Tai et al., bumps 172 or 66, Figure 44).

**Regarding Claim 36,** Tai et al. discloses a system with optical fiber attach capability, the system comprising:

a semiconductor chip package including:

a substrate (substrate 122, Figure 44)

a system on a chip disposed on the substrate (chip 50a, Figure 44);

a photonic integrated circuit with a v-groove in a front side fiber coupling region (PIC, 50c, groove 76, Figure 44);

an organic redistribution layer communicating with the chip and photonic integrated circuit (organic redistribution layer 106, Figure 44); and

an optical fiber attached to the front side fiber coupling region (optical fiber 168, Figure 44),

wherein a mold compound encapsulates the system on a chip, the photonic integrated circuit, and the optical fiber (mold compound 120, Figure 44),

wherein the substrate include a cutout at the attachment of the optical fiber to the photonic integrated circuit (substrate 122, Figure 44).

However, they do not explicitly disclose that the mold compound includes a cutout at the attachment of the optical fiber to the photonic integrated circuit.

Chen et al. discloses a similar device wherein a mold compound encapsulates a chip, a photonic integrated circuit with a groove in a front side, and an optical fiber, said the mold compound including a cutout at the attachment of the optical fiber to the photonic integrated circuit (chip 120, photonic integrated circuit 110c, optical fiber 160c, mold compound 130, Figure 15). Therefore it would have been obvious to one of ordinary skill in the art at the time of the invention to include a cutout in the mold compound at the attachment of the optical fiber to the photonic integrated circuit in Tai et al. in view of Chen et al. in order to simplify the complexity of manufacturing and reduce production cost (Chen et al., Paragraph 38).

Regarding Claim 37, Tai et al. in view of Chen et al. further disclose wherein the apparatus is a die-last wafer- level fanout package (Tai et al., package 601, Figure 44). The language, term, or phrase "die-last wafer-level fanout package," is directed towards the process of making a wafer-level fanout package. It is well settled that "product by process" limitations in claims drawn to structure are directed to the product, per se, no

matter how actually made. In re Hirao, 190 USPQ 15 at 17 (footnote 3). See also, In re Brown, 173 USPQ 685; In re Luck, 177 USPQ 523; In re Fessmann, 180 USPQ 324; In re Avery, 186 USPQ 161; In re Wethheim, 191 USPQ 90 (209 USPQ 554 does not deal with this issue); In re Marosi et al., 218 USPQ 289; and particularly In re Thorpe, 227 USPQ 964, all of which make it clear that it is the patentability of the final product per se which must be determined in a "product by process" claim, and not the patentability of the process, and that an old or obvious product produced by a new method is not patentable as a product, whether claimed in "product by process" claims or otherwise. The above case law further makes clear that applicant has the burden of showing that the method language necessarily produces a structural difference. As such, the language "die-last wafer-level fanout package" only requires a wafer-level fanout package with a die, which does not distinguish the invention from Tai et al. in view of Chen et al., who teaches the structure as claimed.

Regarding Claim 39, Tai et al. in view of Chen et al. further disclose wherein the mold compound comprises an epoxy material (Tai et al., mold compound 120, Figure 44, Paragraph 36).

Regarding Claim 41, Tai et al. in view of Chen et al. further disclose wherein the substrate comprises a plurality of polymer layers (Tai et al., substrate 122, Figure 44, Paragraph 42).

Regarding Claim 44, Tai et al. in view of Chen et al. further disclose wherein a plurality of bumps is attached to the organic redistribution layer (Tai et al., bumps 172 or 66, Figure 44).

Regarding Claim 45, Tai et al. in view of Chen et al. further disclose wherein the plurality of bumps comprises a ball grid array (Tai et al., bumps 172 or 66, Figure 44).

**Claims 12 and 24 are rejected under 35 U.S.C. 103 as being unpatentable over Tai et al. (U.S. 2021/0157052) in view of Chen et al. (U.S. 2020/0271860) as applied to Claim 9 above, further in view of Weng et al. (U.S. 2022/0365297).**

Regarding Claim 12, Tai et al. in view of Chen et al. does not explicitly disclose that the attached fiber is secured by a glob top. However, they disclose the optical fiber is coupled to the package via an adhesive layer (Tai et al., fiber 168, adhesive layer 166, Figure 5, Paragraph 61). Weng et al. discloses a similar device wherein an optical fiber is secured to a PIC that is part of an encapsulated package by a glob top (PIC 50B, optical fiber 168a, glob top 166, Figure 13). Therefore it would have been obvious to one of ordinary skill in the art at the time of the invention to secure the attached fiber by a glob top in Tai et al. in view of Chen et al., further in view of Weng et al. in order enhance coupling efficiency (Weng et al., Paragraph 63).

Regarding Claim 24, Tai et al. in view of Chen et al, further in view of Weng et al. do not explicitly disclose wherein the glob top comprises an epoxy material. Tai et al. disclose that the molding material is an epoxy material (Tai et al., Paragraph 36), Chen et al. discloses attaching an optical fiber using an epoxy material (chip 120, PIC 110,

grooves 112, fiber 160, RDL 140, Figures 14 and 15, Paragraph 17), and Weng et al. discloses that the glob top of a light-transmissive curable gel (Weng et al., Paragraph 63). Therefore it would have been obvious to one of ordinary skill in the art at the time of the invention to form the glob top to comprise an epoxy material in Tai et al in view of Chen et al., further in view of Weng et al., in order to facilitate an optical interface using a suitable optical adhesive to provide optical transparency and mechanical fixation (Chen et al., Paragraph 17). Furthermore it has been held that the selection of a known material based on its suitability for its intended use supported a prima facie obviousness determination in Sinclair & Carroll Co. v. Interchemical Corp., 325 U.S. 327, 65 USPQ 297 (1945), see MPEP 2144.07.

**Claims 38 and 40 are rejected under 35 U.S.C. 103 as being unpatentable over Tai et al. (U.S. 2021/0157052) in view of Chen et al. (U.S. 2020/0271860) as applied to Claim 36 above, further in view of Weng et al. (U.S. 2022/0365297).**

Regarding Claim 38, Tai et al. in view of Chen et al. does not explicitly disclose that the attached fiber is secured by a glob top. However, they disclose the optical fiber is coupled to the package via an adhesive layer (Tai et al., fiber 168, adhesive layer 166, Figure 5, Paragraph 61). Weng et al. discloses a similar device wherein an optical fiber is secured to a PIC that is part of an encapsulated package by a glob top (PIC 50B, optical fiber 168a, glob top 166, Figure 13). Therefore it would have been obvious to one of ordinary skill in the art at the time of the invention to secure the attached fiber by a glob top in Tai et al. in view of Chen et al., further in view of Weng et al. in order enhance coupling efficiency (Weng et al., Paragraph 63).

Regarding Claim 40, Tai et al. in view of Chen et al, further in view of Weng et al. do not explicitly disclose wherein the glob top comprises an epoxy material. Tai et al. disclose that the molding material is an epoxy material (Tai et al., Paragraph 36), Chen et al. discloses attaching an optical fiber using an epoxy material (chip 120, PIC 110, grooves 112, fiber 160, RDL 140, Figures 14 and 15, Paragraph 17), and Weng et al. discloses that the glob top of a light-transmissive curable gel (Weng et al., Paragraph 63). Therefore it would have been obvious to one of ordinary skill in the art at the time of the invention to form the glob top to comprise an epoxy material in Tai et al in view of Chen et al., further in view of Weng et al., in order to facilitate an optical interface using a suitable optical adhesive to provide optical transparency and mechanical fixation (Chen et al., Paragraph 17). Furthermore it has been held that the selection of a known material based on its suitability for its intended use supported a prima facie obviousness determination in Sinclair & Carroll Co. v. Interchemical Corp., 325 U.S. 327, 65 USPQ 297 (1945), see MPEP 2144.07.

**Claims 26 and 27 are rejected under 35 U.S.C. 103 as being unpatentable over Tai et al. (U.S. 2021/0157052) in view of Chen et al. (U.S. 2020/0271860) as applied to Claim 9 above, and further in view of Lim et al. (U.S. 2020/0310052).**

Regarding Claim 26, Tai et al. in view of Chen et al. disclose the limitations of Claim 9 and further disclose the use of microbumps in the package (Tai et al., Paragraph 51) but do not explicitly disclose wherein the system on a chip and the photonic integrated circuit are attached to the substrate with microbumps. Lim et al. discloses a similar device wherein a chip and PIC are attached to a substrate with bumps (Lim et al., chip 420, PIC

406, substrate 402, bumps 422, Figures 4g and 4h). Therefore it would have been obvious to one of ordinary skill in the art at the time of the invention to attach the SOC and PIC to the substrate at a fine, micro pitch using microbumps in Tai et al. in view of Chen et al., further in view of Lim et al. in order to form a device with lower cost and greater speed (Lim et al., Paragraphs 45 and 46).

Regarding Claim 27, Tai et al. in view of Chen et al., further in view of Lim et al. further disclose wherein the microbumps are secured by an underfill (Lim et al., underfill 426, Figure 4h).

**Claims 42 and 43 are rejected under 35 U.S.C. 103 as being unpatentable over Tai et al. (U.S. 2021/0157052) in view of Chen et al. (U.S. 2020/0271860) as applied to Claim 36 above, and further in view of Lim et al. (U.S. 2020/0310052).**

Regarding Claim 42, Tai et al. in view of Chen et al. disclose the limitations of Claim 9 and further disclose the use of microbumps in the package (Tai et al., Paragraph 51) but do not explicitly disclose wherein the system on a chip and the photonic integrated circuit are attached to the substrate with microbumps. Lim et al. discloses a similar device wherein a chip and PIC are attached to a substrate with bumps (Lim et al., chip 420, PIC 406, substrate 402, bumps 422, Figures 4g and 4h). Therefore it would have been obvious to one of ordinary skill in the art at the time of the invention to attach the SOC and PIC to the substrate at a fine, micro pitch using microbumps in Tai et al. in view of Chen et al., further in view of Lim et al. in order to form a device with lower cost and greater speed (Lim et al., Paragraphs 45 and 46).

Regarding Claim 43, Tai et al. in view of Chen et al., further in view of Lim et al. further disclose wherein the microbumps are secured by an underfill (Lim et al., underfill 426, Figure 4h).

## *Response to Arguments*

Applicant's arguments with respect to claims 9 and 36 have been considered but are moot because the new ground of rejection does not rely on any reference applied in the prior rejection of record for any teaching or matter specifically challenged in the argument.

## *Conclusion*

The prior art made of record and not relied upon is considered pertinent to applicant's disclosure.

Huang et al. (U.S. 11,222,867) discloses a device package wherein a PIC with a v-groove is mounted on a substrate and encapsulated with another chip.

Yu et al. (U.S. 10,866,373) discloses a device package wherein a die with a v-groove is mounted on a substrate and encapsulated with another chip.

Wang et al. (U.S. 10,162,139) discloses a device package wherein a die with a v-groove is mounted on a substrate and encapsulated with another chip.

Miura et al. (U.S. 6,170,996) discloses a device package wherein a die with a v-groove is mounted on a substrate and encapsulated with another chip.

Applicant's amendment necessitated the new ground(s) of rejection presented in this Office action. Accordingly, **THIS ACTION IS MADE FINAL**. See MPEP § 706.07(a). Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).

A shortened statutory period for reply to this final action is set to expire THREE MONTHS from the mailing date of this action. In the event a first reply is filed within TWO MONTHS of the mailing date of this final action and the advisory action is not mailed until after the end of the THREE-MONTH shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any extension fee pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the date of this final action.

Any inquiry concerning this communication or earlier communications from the Examiner should be directed to Abbigale Boyle whose telephone number is 571-270-7919. The Examiner can normally be reached from 11 A.M to 7 P.M., Monday through Friday.

If attempts to reach the Examiner by telephone are unsuccessful, the Examiner's supervisor, Zandra Smith, can be reached at 571-272-2429. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197

(toll-free). If you would like assistance form a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

Abbigale Boyle

Examiner, Art Unit 2816

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/DALE E PAGE/
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